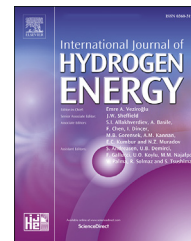


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Catalytic hydrogen evolution reaction on surfaces of metal-nanoparticle-coated zinc-based oxides by first-principles calculations

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HIGHLIGHTS

- The HER performance of NPs-coated oxide were studied by first principle calculation.
- The NPs surface of Au/ZnO structure shows the lowest adsorption free energy.
- Pt cluster is the preferred surface for hydrogen spillover according to Bader charge.
- Volmer-Heyrovsky mechanism on NPs surface of Pt/ZnO has the best HER performance.

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ABSTRACT

Using first-principles calculations, we investigate the hydrogen evolution reaction (HER) on the noble-metal-nanoparticle-coated zinc or zinc-tin oxide surfaces. Among the six catalytic structures, the adsorption free energy of Au/ZnO structure is closest to zero. The relative energy diagrams reveal that Volmer-Heyrovsky mechanism on the NP (oxide) reactive site dominants the HER process on NP/ZTO (Pt/ZnO and Au/ZnO) structures. However, comparing the adsorption free energy and primary energy barrier of the prefer path of each structure, the Pt/ZnO structure shows the best performance for the HER process.

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Introduction

The demand for energy grows exponentially, but the use of fossil fuel and coal produce air pollution and can lead to global warming [1–4]. Hydrogen energy has thus been considered as

a potential energy source of next generation, owing to its high energy density, high conversion efficiency, and environmental friendly combustion product [5–10].

Zinc oxide (ZnO) can be a prominent photocatalyst material owing to its high electron mobility, low cost, and non-

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toxic. These desired properties benefiting ZnO in photocatalytic hydrogen evolution [11–13]. For example, the ZnO–CdS core–shell nanorods structure shows stable and high hydrogen evolution ability [14]. The Au decorated hollow ZnO@ZnS heterostructure also exhibits great hydrogen evolution efficiency due to the enhanced separation and migration of charge carriers [15]. However, the main challenges of ZnO catalytic structure are the high electron-hole recombination rate and the wide bandgap that only absorbs UV light [16,17]. An effective solution to enhance photocatalytic efficiency of ZnO structure is the coating of noble metal (Pt, Au or Ag) nanoparticles as cocatalyst [18–22]. The Pt nanoparticle-coated ZnO (NP/ZnO) nanorod largely improves the photoelectrochemical activity of ZnO structure [23], and the Au NP/ZnO structure also shows better water splitting performance [24]. The noble metal NPs act as electron scavengers, delaying the recombination process and acting as H₂ evolution site [25,26]. Another approach for improving efficiency is the substitution of Zn²⁺ ions with higher valence cations such as Sn⁴⁺ that can act as efficient shallow donors, whereas Zn₂SnO₄ (ZTO) has high electron mobility and chemical stability [27], leading to its photocatalytic applications [28]. Pure ZTO, Pt/ZTO and Au/ZTO are capable of performing HER while the Pt/ZTO shows the best performance [29].

Comparing with other noble metals, Pt shows better catalytic efficiency, which can be attributed to the hydrogen spillover on Pt surface [30,31]. Hydrogen spillover has been well observed on Pt and Pd nanoparticle surfaces [32,33], which starts with the dissociation of hydrogen molecules on the nanoparticle and then the migration of hydrogen atom to the supported oxide. It has been applied in improving the photoefficiency of catalytic materials [34]. The migration of hydrogen atom provides two benefits to improve the reaction efficiency—(1) It enhances the separation of hydrogen adsorbing site and desorbing site, thus provide more reactive sites for hydrogen evolution reaction. (2) It increases the chance for two adsorbed hydrogens to meet each other for forming a molecule [35,36]. Understanding the HER mechanism on noble metal NP/ZnO (ZTO) structure is key for further improving its performance, but the related study is very limited. First principle calculation has been applied in catalytic field to study the microscopic perspective of hydrogen evolution reaction [37,38]. The calculation results reveal the activity, energy barrier, and other material properties that enable us to understand the reaction mechanism in atomic scale [39,40].

In this work, HER steps had been investigated by first-principles calculations. Both nanoparticle and oxide surfaces are considered as possible reactive sites. Deformation charge diagram and Bader charge were calculated for charge transfer analysis. Free energy diagrams were plotted in order to compare HER performance of each structure. D-band center calculations were performed to evaluate the HER activity. Climb image nudge-elastic band method was applied to obtain the energy barrier of each HER step and established the complete reaction mechanism of each structure.

Computational details

Computational model

To explore NP/ZTO and NP/ZnO heterostructures, we first established $2 \times 2 \times 1$ ZTO (001) and $4 \times 4 \times 2$ ZnO (002) supercells and then optimized these structures, where the lowest five atomic layers were constrained during the optimization. A 15 Å vacuum gap from the surface was added. Based on Watari's report [41], the least number of atoms to form cuboctahedral or icosahedral structure is 13. In this work, a nanoparticle composed of 13 noble metal atoms (Pt₁₃, Au₁₃ and Ag₁₃) was optimized, and then the NP₁₃ was adsorbed on the optimized ZTO (001) and ZnO(002) surfaces along the z-direction. The NP/ZTO and NP/ZnO structures have lattice parameters of 9 and 8 Å along the x and y directions, respectively.

Computational detail

First-principles calculations were performed using Vienna ab initio simulation package (VASP) [42,43]. The generalized gradient approximation with Perdew-Burke-Ernzerhof function was adopted to describe the exchange-correlation interaction [44,45]. The cutoff energy was set to 520 eV, and the convergence tolerances for the residual force and energy on each atom during the structural relaxation were set to 0.05 eV and 10^{−5} eV, respectively. The Brillion zone integral was generated by the gamma method. Climbing image-nudging elastic band method [46] was used for transition state and energy barrier calculations.

Results and discussion

Optimized H-adsorbed NP/ZTO and NP/ZnO structures

The bond formation is supported by the projected density of states (PDOS) of hydrogen-adsorbed NP/ZTO and NP/ZnO structures, shown in Table S2. The overlaps near the Fermi level of the H-1s orbital and the Pt-, Au-, or Ag-d orbital verify the chemical bond between the hydrogen atom and the adsorbed atom on NP surface of ZTO and ZnO structures. Finally, the overlaps near the Fermi level of the H-1s orbital and the O-2p orbital confirm the bond between H and O atoms.

Charge transfer analysis

To further explore the bonding states between hydrogen and noble metal or oxygen atoms, deformation charge densities and Bader charges were calculated. The deformation charge densities are shown in Fig. 1(a–k). It is confirmed that the adsorbed hydrogen atom is not only spatially shared by two Pt or Au atoms on a NP surface but also with the same amount of charge transfer between them. This also stays true for a hydrogen atom shared by three Ag atoms.

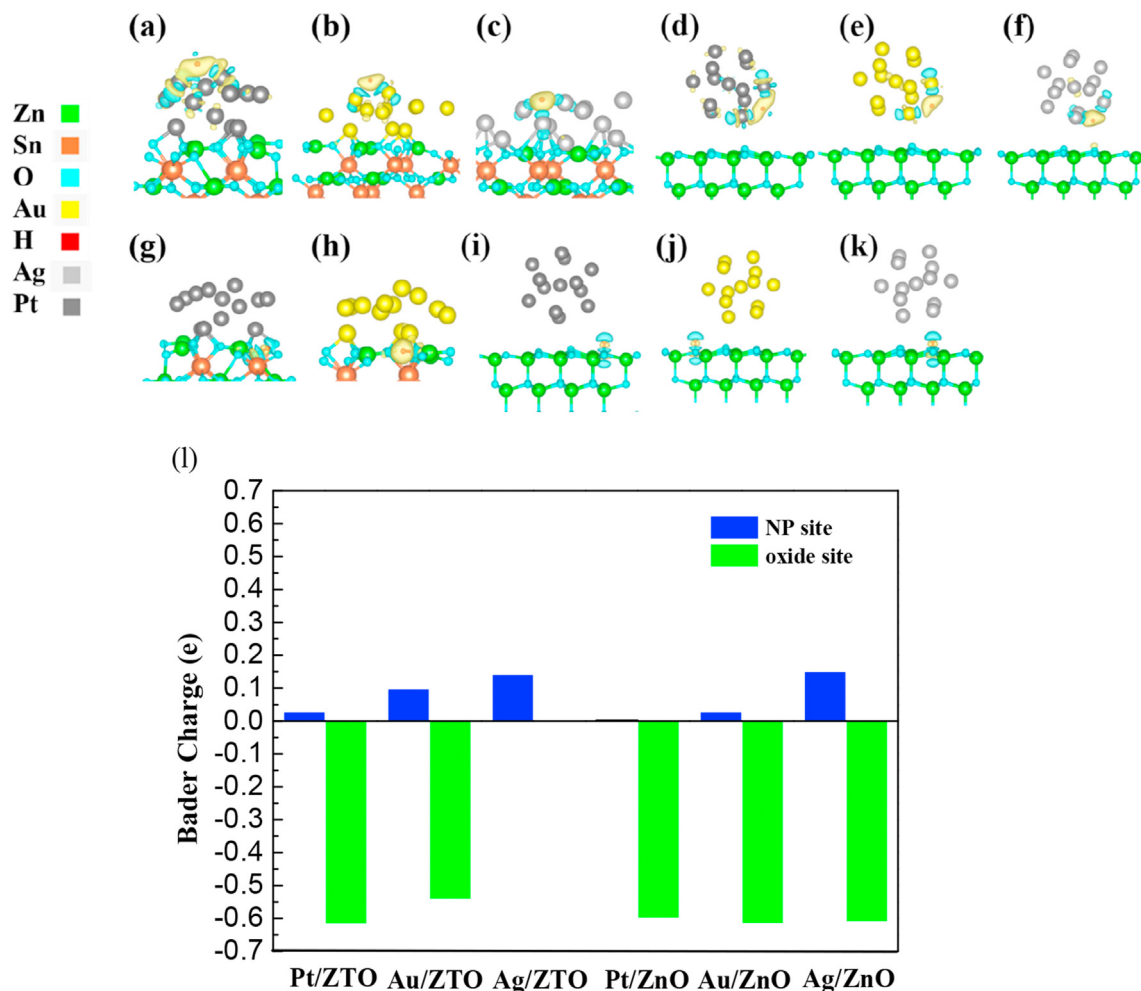


Fig. 1 – Side views of the deformation charge density of interface between the adsorbed H atom and (a) the Pt reactive site of Pt/ZTO structure, (b) the Au reactive site of Au/ZTO structure, (c) the Ag reactive site of Ag/ZTO structure, (d) the Pt reactive site of Pt/ZnO structure, (e) the Au reactive site of Au/ZnO structure, (f) the Ag reactive site of Ag/ZnO structure, (g) the oxygen reactive site of Pt/ZTO structure, (h) the oxygen reactive site of Au/ZTO structure, (i) the oxygen reactive site of Pt/ZnO structure, (j) the oxygen reactive site of Au/ZnO structure, and (k) the oxygen reactive site of Ag/ZnO structure. The yellow (blue) part represents gaining (losing) of charge. (l) Net Bader Charge values for hydrogen atoms adsorbed on reactive sites of NP/ZTO and NP/ZnO structures. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Fig. 1l shows the net Bader charges of the adsorbed hydrogen atom. The H-atom adsorbed on a NP gains charge from the NP, while it adsorbed on an oxide loses charge to the oxygen atom. A H-atom tends to gain more charge from a Ag NP, while gaining less charges from the Pt NP after the adsorption process. The net Bader charge is near zero for Pt/ZnO case. Since hydrogen spillover strongly depends on the charge losing of the adsorbed hydrogen, the hydrogen on oxide surface is more likely to migrate through spillover with respect to hydrogen on noble metal surface. Among all three noble metals, hydrogen spillover should only be observed on Pt nanoparticles which lose near zero charge to the adsorbed hydrogen.

Adsorption energy and photocatalytic activity

The adsorption energy of hydrogen was calculated according to following equation:

$$\Delta E_{\text{ad}} = E_{(\text{NP/oxide} + \text{H}^*)} - \frac{1}{2}E_{(\text{NP/oxide})} - E_{(\text{H}_2)} \quad (1)$$

where $E_{(\text{NP/oxide} + \text{H}^*)}$ is the energy of the NP/oxide structure with an adsorbed hydrogen. $E_{(\text{NP/oxide})}$ is the energy of isolated NP/ZTO or NP/ZnO structures, and $E_{(\text{H}_2)}$ is the energy of a hydrogen molecule.

The adsorption free energy of hydrogen was calculated using the following equation:

$$\Delta G_{\text{H}}^* = E_{\text{ads}} + \Delta E_{\text{ZPE}} - T\Delta S \quad (2)$$

where E_{ads} is the adsorption energy of hydrogen. ΔE_{ZPE} is the zero-point energy difference between the adsorbed hydrogen and the hydrogen in the gas phase, and $T\Delta S$ is the entropy change during hydrogen changes from the gas phase to the adsorbed state. The value of ΔE_{ZPE} can be neglected because

the vibrational entropy in an adsorbed state is much than the gas phase. The $T\Delta S$ value at room temperature (298.15 K) is ~ 0.2 eV for each Hydrogen atom. Nørskov [47] proposed that $\Delta E_{ZPE} - T\Delta S$ is ~ 0.24 eV for metal catalyst.

Based on Sabatier principle [48], the free energy change of H adsorption (ΔG_H^*) is a determinant parameter for evaluating the HER efficiency of a catalytic material. If ΔG_H^* is too high, the adsorption of H is unstable and thus prevent the formation of H^* transition state. If ΔG_H^* is too low, the hydrogen molecule will be unable to leave the catalytic surface. The ideal catalytic material should have an ΔG_H^* near zero. Fig. 2a shows the adsorption free energy diagrams of hydrogen on the NP surface of NP/ZTO and NP/ZnO structures. The NP reactive site of Au/ZnO structure has the smallest $|\Delta G_H^*|$ (~ 0.01 eV). Fig. 2b shows the oxide surface version of the adsorption free diagrams. For oxide reactive site, Au/ZnO structure also shows the smallest $|\Delta G_H^*|$ (~ 0.02 eV). The results indicate that the Au/ZnO structure is thermodynamically the best catalyst for HER process. In contrast, the adsorption free energy of NP reactive site of Ag/ZTO is as high as 6.14 eV. With such high ΔG_H^* , the Ag/ZTO structure should show the worst catalytic activity among all. These simulation results supported the superior performance of Au/ZnO compared to Ag/ZnO in photocatalytic application [49]. For the Pt/ZnO structure, it has the most negative ΔG_H^* value (-0.52 eV) on NP reactive site and near zero $|\Delta G_H^*|$ value (-0.06 eV) on the oxide reactive site, indicating that the Pt surface has a strong ability to catch proton in water while ZnO surface is a suitable reactive site for HER. Based on this result, hydrogen would prefer to absorb on NP site first then desorbed as hydrogen molecule on oxide surface.

Bonding state analysis

The d-band center theory proposed by Nørskov [50,51] suggested that the d-band center can be regarded as an indicator for determining adsorption energy and catalytic activity. The closer d-band stands to the Fermi level, the less valence bands are occupied by the antibonding states, leading to a stronger bonding strength. The d-band center energy (ϵ_d) is defined as

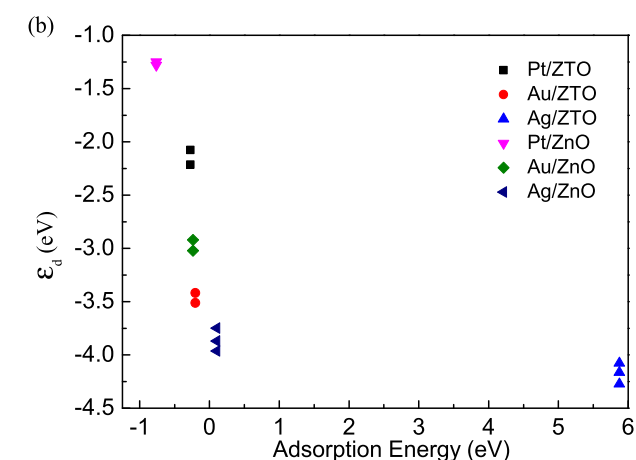


Fig. 3 – D-band center vs E_{ad} plot of NPs on NP/ZTO and NP/ZnO structures.

$$\epsilon_d = \frac{\int_{-\infty}^{\infty} \rho(\epsilon) \epsilon d\epsilon}{\int_{-\infty}^{\infty} \rho(\epsilon) d\epsilon} \quad (3)$$

where ρ is the densities of states and ϵ is the energy.

Since hydrogen tends to be shared by multiple atoms, the d-band centers of all adsorbed atoms of NP/ZTO and NP/ZnO structures are plotted, as a function of the adsorption free energy (E_{ad}), in Fig. 3. The d-band center is inversely correlated with the adsorption energy, in consistent with the d-band theory. The closer distance between the d-band center of Pt/ZnO structure and the Fermi level provides an explanation for its relative negative adsorption free energy.

Prefer reaction path analysis

The formation of hydrogen molecules can be taken by two different routes- Volmer-Tafel or Vomer-Heyrovsky steps. Both routes start with Volmer step – an aqueous proton adsorbed on the surface of catalytic materials. Then the adsorbed proton can either combine with another adsorbed

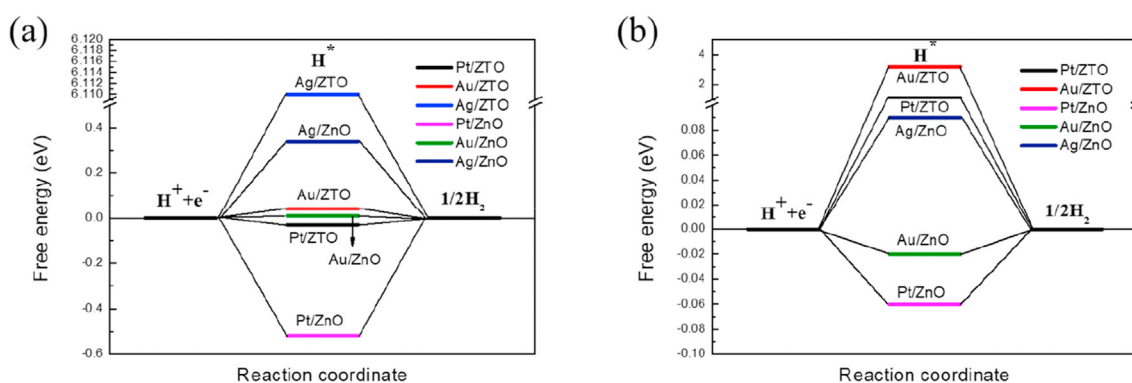


Fig. 2 – Free energy diagram of hydrogen evolution on (a) the NP reactive site of NP/ZTO and NP/ZnO structures, and (b) the oxide reactive site of NP/ZTO and NP/ZnO structures.

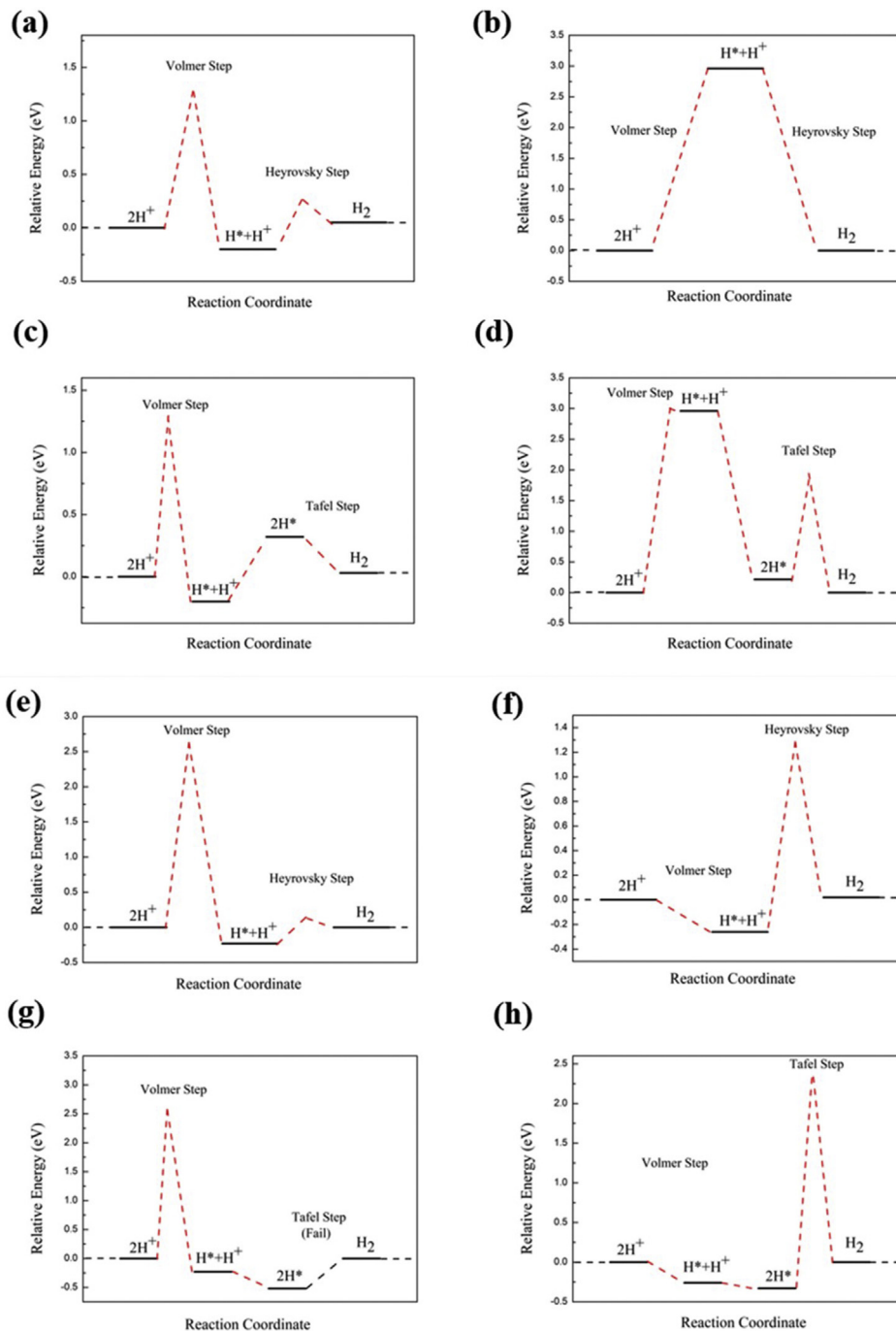


Fig. 4 – Relative energy diagram of (a) Volmer-Heyrovsky step on the NP reactive site of Au/ZTO structure, (b) Volmer-Heyrovsky step on the oxygen reactive site of Au/ZTO structure, (c) Volmer-Tafel step on the NP reactive site of Au/ZTO structure, (d) Volmer-Tafel step on the oxygen reactive site of Au/ZTO structure, (e) Volmer-Heyrovsky step on the NP reactive site of Au/ZnO structure, (f) Volmer-Heyrovsky step on the oxygen reactive site of Au/ZnO structure, (g) Volmer-Tafel step on the NP reactive site of Au/ZnO structure, and (h) Volmer-Tafel step on the oxygen reactive site of Au/ZnO structure.

proton (Tafel step) or go through a second proton/electron transfer to form molecular hydrogen (Heyrovsky step) [52,53].



Based on the calculated energy barriers and optimized structures, the relative energy diagrams for hydrogen evolution reaction on all NP/ZTO and NP/ZnO structures are shown in Fig. 4, where the relative energy (barrier) of each step is represented with black bar (red peak). If the optimization is

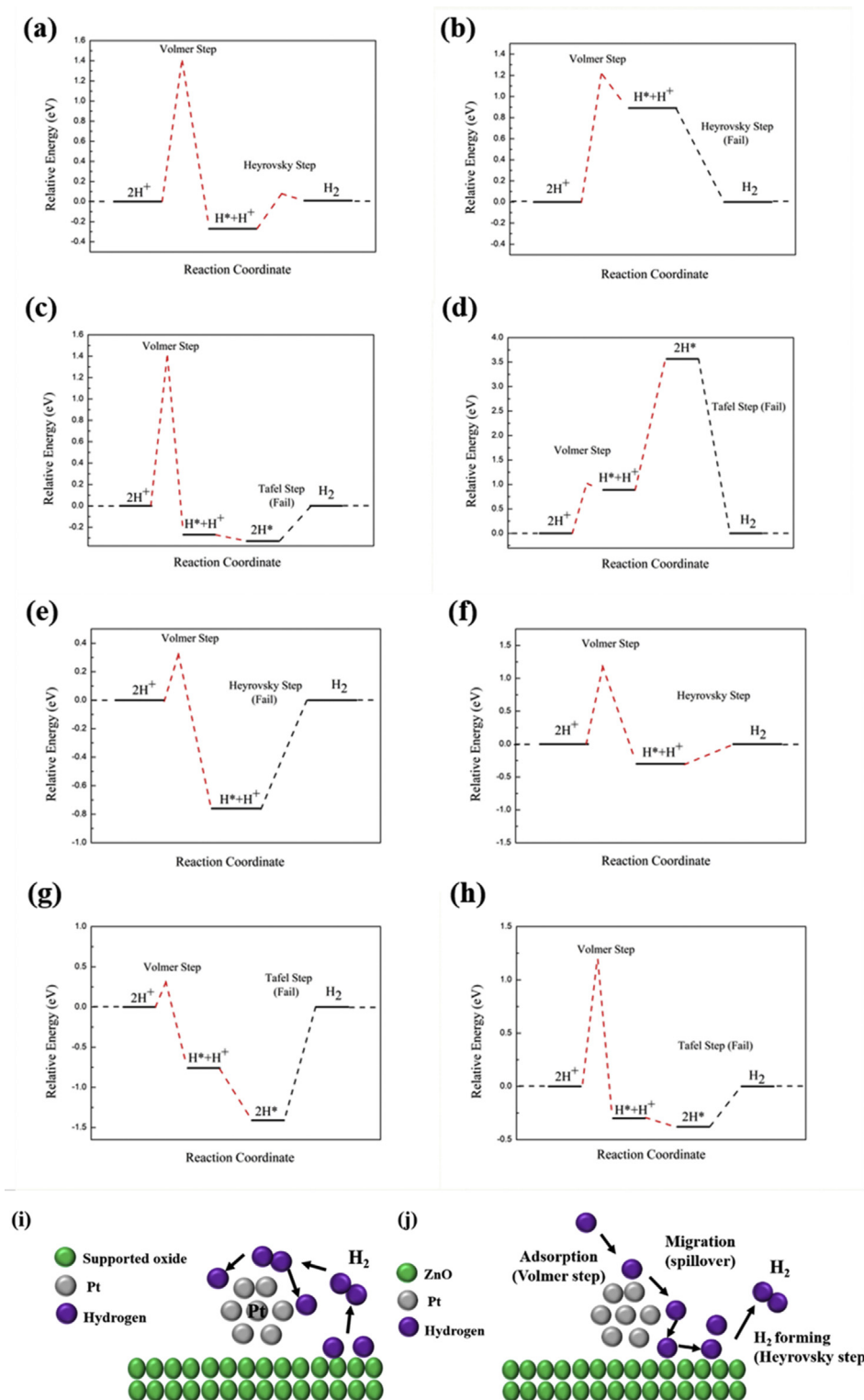


Fig. 5 – Relative energy diagram of (a) Volmer-Heyrovsky step on the NP reactive site of Pt/ZTO structure, (b) Volmer-Heyrovsky step on the oxygen reactive site of Pt/ZTO structure, (c) Volmer-Tafel step on the NP reactive site of Pt/ZTO structure, (d) Volmer-Tafel step on the oxygen reactive site of Pt/ZTO structure, (e) Volmer-Heyrovsky step on the NP reactive site of Pt/ZnO structure, (f) Volmer-Heyrovsky step on the oxygen reactive site of Pt/ZnO structure, (g) Volmer-Tafel step on the NP reactive site of Pt/ZnO structure, and (h) Volmer-Tafel step on the oxygen reactive site of Pt/ZnO structure. (i) Schematic illustration of re-adsorption and spillover process on the Pt NP surface. (j) Schematic illustration of hydrogen evolution reaction mechanism on the Pt/ZnO structure.

Table 1 – The adsorption free energy and primary energy barrier of the prefer paths of NP/ZTO and NP/ZnO structures.

	Pt/ZTO	Au/ZTO	Ag/ZTO	Pt/ZnO	Au/ZnO	Ag/ZnO
Reactive site	Pt NPs	Au NPs	Ag NPs	ZnO surface	ZnO surface	Ag NPs & ZnO surface
E_{ad} (eV)	0.03	0.04	6.11	−0.06	−0.02	0.34 or 0.09
$E_{barrier}$ (eV)	1.41	1.32	7.38	1.23	1.52	1.14 or 1.22

failed at any reaction step, the step will be labeled with a fail mark. Four possible reaction routes are proposed based on the optimized NP/ZTO and NP/ZnO structures: The Volmer-Heyrovsky step on the NP reactive site, the Volmer-Heyrovsky step on the oxygen reactive site, the Volmer-Tafel step on the NP reactive site, and the Volmer-Tafel step on the oxygen reactive site.

Fig. 4(a–d) show the relative energy diagrams of the Au/ZTO structure. HER can be successfully performed on both Au and ZTO surfaces. The E_{ad} and $E_{barrier}$ of each possible reaction path are compared to find out the prefer path. HER route on the ZTO surface is out of option due to its high and positive adsorption free energy (3.2 eV). Then by comparing the energy barrier of Heyrovsky step (0.4 eV) and energy of $H^* \rightarrow 2H^*$ state (0.52 eV) in Tafel step, the Volmer-Heyrovsky step on Au surface turns out to be more favorable due to its lower energy barrier. Fig. 4(e–h) shows the relative energy diagrams of the Au/ZnO structure. To find out the prefer path, we compare the energy barrier of Volmer step on Au reactive site (2.74 eV) with Tafel step (2.3 eV) and Heyrovsky step (1.52 eV) on ZnO surface. The result shows that Volmer Heyrovsky step on ZnO surface is more favorable for HER.

The relative energy diagrams of Ag/ZTO structure are showed in Fig. S1(a–b). After optimization, hydrogen cannot adsorb on the ZTO surface of the Ag/ZTO structure, the Volmer step which involve the adsorption of hydrogen is unlikely to happen. Therefore, HER cannot occur on the ZTO surface of the Ag/ZTO structure. On the other hand, the high adsorption free energy (6.14 eV) and energy barrier (7.38 eV) make it thermodynamically and kinetically unfavorable for HER on the Ag reactive site of the Ag/ZTO structure. Fig. S1(c–f) shows the relative energy diagrams of the Ag/ZnO structure. Hydrogen prefers to adsorb on the Ag reactive site rather than on the ZnO surface because the former has less adsorption free energy (0.09 eV) than the latter. However, comparing the energy barrier of the Volmer step on the Ag reactive site (1.14 eV) with those of the Volmer (1.22 eV) and Tafel (2.05 eV) steps on the ZnO surface, the Volmer-Heyrovsky step on the ZnO surface is more favorable. Consequently, it is hard to tell which one is more preferred for HER, since the Volmer-Heyrovsky step on the ZnO surface is kinetically favorable while the Volmer-Heyrovsky step on the Ag reactive site is thermodynamically favorable.

Fig. 5(a–h) show the relative energy diagrams of Pt/ZTO and Pt/ZnO structures. During the structural optimization, a unique process was observed on both structures. Fig. 5i shows the schematic illustration of this process on the Pt/ZTO structure. First, hydrogen molecule formed by the Volmer-Tafel step or the Volmer-Heyrovsky step on the ZTO surface of the Pt/ZTO structure. Then it re-adsorbs on the surface of the Pt_{13} cluster. Finally, it spills over into two adsorbed

hydrogens. Similar process is also observed during the optimization of the Vomer-Tafel step on the Pt surface of the Pt/ZnO structure. The phenomenon meets with the description of hydrogen spillover in previous studies [35,36]. Since the calculations of HER on many different paths end up with two adsorbed protons due to hydrogen spillover, only the Volmer-Heyrovsky step on the Pt reactive site is successfully performed for the Pt/ZTO structure. For the Pt/ZnO structure, only the Volmer-Heyrovsky step on the ZnO surface was successfully performed.

To estimate the best catalytic structure for hydrogen evolution reaction, the primary energy barrier and the adsorption free energy of a prefer path of NP/ZTO and NP/ZnO structures are summarized on Table 1. Both Ag/ZTO and Ag/ZnO structures are out of options for their relative high adsorption free energies. Comparing the energy barriers of the four remaining structures, the Pt/ZnO structure is the best catalytic material for HER owing to its low energy barrier. The simulation results reveal that Pt/ZnO is kinetically favorable among all. This is in line with experimental observation [54].

Based on our calculations, we established the HER mechanism on the Pt/ZnO structure in Fig. 5j. First, the aqueous proton adsorbed on the Pt surface due to the strong binding strength of the Pt reactive site. Then the adsorbed proton migrates to the ZnO surface through spillover effect. Eventually, molecular hydrogen is formed through the Heyrovsky step reaction on the ZnO surface.

Conclusion

HER on Noble-metal-NPs-coated metal oxides are investigated from first principles. The optimized structures show that hydrogen is unstable to adsorb on the ZTO surface of the Ag/ZTO structure. PDOS and deformation charge density analyses confirm that the adsorbed hydrogen tends to be shared by multiple noble metal atoms. Bader charge calculation reveals that hydrogen spillover is more likely to occurs on the Pt surface than on the Au or Ag surfaces. Based on the adsorption free energy, the Au/ZnO structure exhibits the best HER activity among all NP/ZTO and NP/ZnO structures. Calculated d-band center energy corresponds with the adsorption energy, explaining the relative negative E_{ad} value of the Pt/ZnO structure. From relative energy diagrams, the HER prefer path for Pt/ZTO and Au/ZTO (Pt/ZnO and Au/ZnO) structures is the Volmer-Heyrovsky step on the NP (ZnO) reactive site. The Ag/ZTO structure is thermodynamically unfavorable for HER due to its high adsorption free energy. For the Ag/ZnO structure, it is hard to tell whether the Volmer-Heyrovsky step on the Ag reactive site or on the ZnO surface is the prefer path because the previous is kinetically favorable and the latter is

thermodynamically favorable. Hydrogen spillover occurs on the Pt surface of both Pt/ZnO and Pt/ZTO structures. Based on the E_{ad} and $E_{barrier}$ of the prefer path of NP/ZTO and NP/ZnO structures, the Pt/ZnO structure is the best catalytic structure for HER. Finally, the complete HER mechanism of the Pt/ZnO structure is found from calculation results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijhydene.2021.09.173>.

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